

AI2002 – Artificial Intelligence

Quiz – 3B

Name	
Roll. No.	
Section	

CLO3: Design basic AI-based solutions for real-world problems using appropriate techniques such as reasoning, tracking, or robot localization.

Scenario: Four milestones $\{M1, M2, M3, M4\}$ must be assigned to phases. Domains $D = \{\text{Phase 1 (P1), Phase 2 (P2), Phase 3 (P3)}\}$. Constraints: (M1 before M2), $(M2 \neq M3)$, (M3 before M4), $(M1 \neq M3)$.

(Note: $P1 < P2 < P3$)

Part 1: Forward Checking (3 Marks)

Assume milestone $M1 = P2$. Apply Forward Checking to update the domains.

Milestone	Initial Domain	After M1 = P2 (Forward Checking)
M2	{P1, P2, P3}	
M3	{P1, P2, P3}	
M4	{P1, P2, P3}	

Part 2: MRV & Selection (4 Marks)

Using the result from Part 1: select the next milestone using MRV, assign it the earliest possible phase, and apply Forward Checking again.

Selected Milestone: _____ Phase Assigned: _____

Remaining Milestone	Domain from Part 1	New Domain (After FC)

Does this branch result in a failure? [Yes / No]

Part 3: Forward Checking vs. Arc Consistency (3 Marks)

Consider a separate milestone CSP with Variables $\{X, Y, Z\}$, Domains $\{P1, P2, P3\}$, and Constraints (X before Y), (Y before Z).

1. If we assign $X = P1$, what are the domains of Y and Z using Forward Checking?

$D(Y) = \{ \text{_____} \}$ $D(Z) = \{ \text{_____} \}$

2. Now apply Arc Consistency to the constraint (Y before Z). Does it remove any additional value from the domain of Y?

Result: _____

3. Which method gives a smaller domain for Y after propagation? _____